Lecture 14

Operators on Hilbert Space

This final lecture carries the operator theory of Part I—adjoints, the spectral theorem, the operator classes of Lecture 9—into the infinite-dimensional Hilbert spaces of Lecture 13, where it becomes the mathematical language of quantum mechanics. Two things change in infinite dimensions. First, eigenvalues no longer tell the whole story: an operator can be "non-invertible in the limit" without having any eigenvector, giving a *continuous spectrum*. Second, the operators of physics—position, momentum, energy—are *unbounded*, defined only on dense domains, and treating them rigorously is essential. We build operator norms and adjoints, classify the spectrum, meet position and momentum and their canonical commutation relation, and state the spectral theorem in the form quantum mechanics needs. The course closes with the dictionary between this linear algebra and the postulates of quantum theory.

Remark 14.1 (Standing scope for Lecture 14). Throughout, $\mathcal{H}$ is a complex Hilbert space and $\mathcal{B}(\mathcal{H})$ denotes its bounded operators. The algebraic notation $\mathcal{L}(V,W)$ retains its earlier meaning. An unbounded operator is a linear expression together with a specified dense domain; adjoints, products, commutators, and functions of such operators retain domain conditions.

Learning objectives

By the end of this lecture you will be able to:

- Use the operator norm and adjoint for bounded operators, carrying over the Lecture 9 classes.
- Classify the spectrum (point / continuous / residual) as the generalization of eigenvalues.
- Explain why position and momentum are unbounded and why $[\hat{x},\hat{p}] = i\hbar$ has no finite model.
- State the spectral theorem (compact and general self-adjoint) and read off the QM dictionary.

Remark 14.2 (Core, bridge, and extension route). The assessable core is $\mathcal{B}(\mathcal{H})$, adjoints, the three spectral parts, one shift, direct unboundedness, common-domain CCR and trace obstruction, boundary conditions, the qualified compact theorem, a domain-aware PVM theorem, and the pure/projective/closed quantum dictionary. Resolvents, compact-resolvent physics, Stone's theorem, multiplication/free-particle models, and finite ideal measurement calculations are bridges. Repeated ladder algebra, kernel proofs, squeezed/coherent vistas, and general measurement/open-system developments are extensions.

1 Bounded operators and the adjoint

Definition 14.3 (Bounded operator; operator norm). A linear operator $T\colon \mathcal{H} \to \mathcal{H}$ is *bounded* if

$$\|T\| = \sup_{v\neq 0} \frac{\|Tv\|}{\|v\|} < \infty.$$

Bounded is equivalent to continuous. The bounded operators form a Banach space $\mathcal{B}(\mathcal{H})$ under the operator norm.

Definition 14.4 (Adjoint). The *adjoint* $T^\dagger$ of a bounded operator is the unique bounded operator with $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ for all u, v (it exists by the Riesz theorem). As in Lecture 9, T is *self-adjoint* if $T^\dagger = T$, *unitary* if $T^\dagger T = TT^\dagger = \mathrm{id}$, and *normal* if $T^\dagger T = TT^\dagger$.

These classes mean exactly what they did in finite dimensions, and the same proofs show self-adjoint operators have real expectation values and unitary operators preserve the inner product. What is new is everything to do with the spectrum.

Example 14.5 (Operator norms). The identity has $\|\mathrm{id}\| = 1$, and any nonzero *orthogonal* projection has norm 1 (an oblique idempotent can have norm larger than 1). A diagonal operator on ℓ^2 with entries d_n has norm $\sup_n |d_n|$, finite exactly when the d_n are bounded. Multiplication by $m \in L^\infty$ on L^2 has norm $\operatorname{ess\,sup}|m|$; a value changed on a null set does not change the operator. ◁

Example 14.6 (Bounded versus unbounded). Multiplication by $\sin x$ on $L^2(\mathbb{R})$ is bounded ($\|M\| = 1$); multiplication by x is not. Indeed the normalized functions $\psi_n = \mathbf{1}_{[n,n+1]}$ satisfy $\|\psi_n\| = 1$ and

$$\|\hat{x}\psi_n\|^2 = \int_n^{n+1} x^2\, dx = n^2 + n + \tfrac{1}{3} \longrightarrow \infty.$$

Boundedness is exactly a uniform bound on this ratio—which observables like position lack. ◁

Example 14.7 (Adjoints). The adjoint of multiplication by m is multiplication by $\overline{m}$, so a real-valued multiplier is self-adjoint. The right shift S on ℓ^2 has adjoint the left shift $S^\dagger(x_1, x_2, \dots) = (x_2, x_3, \dots)$, with $S^\dagger S = \mathrm{id}$ but $SS^\dagger \neq \mathrm{id}$—an isometry that is not unitary, impossible in finite dimensions. ◁

Example 14.8 (Self-adjoint, unitary, normal). Multiplication by a real essentially bounded function is self-adjoint; multiplication by $e^{i\theta(x)}$ is unitary; the Fourier transform is unitary. Hamiltonian, position, and momentum *observables* are chosen self-adjoint realizations of their formal expressions on specified domains—the operators whose spectra are physically observable values. ◁

Example 14.9 (A norm bound). For any bounded T, $|\langle u, Tv\rangle| \le \|u\|\,\|T\|\,\|v\|$ by Cauchy–Schwarz; one also has $\|T^\dagger\| = \|T\|$ and (stated without proof) $\|T^\dagger T\| = \|T\|^2$, the C^*-identity. These make $\mathcal{B}(\mathcal{H})$ a C^*-algebra, the abstract setting for quantum observables. ◁

2 The spectrum

Definition 14.10 (Spectrum). For $T \in \mathcal{B}(\mathcal{H})$, the *spectrum* is

$$\operatorname{spec}(T) = \{\lambda \in \mathbb{C} : T - \lambda\,\mathrm{id} \text{ has no bounded everywhere inverse in } \mathcal{B}(\mathcal{H})\}.$$

Writing $A = T - \lambda\,\mathrm{id}$, the disjoint cases are: *point spectrum* when $\operatorname{null} A \neq \{0\}$; *continuous spectrum* when A is injective with dense range but no bounded everywhere inverse; and *residual spectrum* when A is injective with non-dense range.

In finite dimensions "not invertible" means "has a kernel," so the spectrum is exactly the eigenvalues. In infinite dimensions an injective operator may have dense nonclosed range (continuous spectrum) or non-dense range (residual spectrum). For a self-adjoint operator the spectrum is real; for a unitary operator it lies on the unit circle—the infinite-dimensional shadows of Lecture 9.

Example 14.11 (Continuous spectrum). Multiplication by x on $L^2[0,1]$ has *no* eigenvectors: $x\psi(x) = \lambda\psi(x)$ forces $\psi = 0$ almost everywhere. Yet $\hat{x} - \lambda\,\mathrm{id}$ (multiplication by $x - \lambda$) is not boundedly invertible for any $\lambda \in [0,1]$, since $1/(x-\lambda)$ is unbounded there. Its range is dense: if g is orthogonal to that range, then $(x-\lambda)g = 0$ almost everywhere, and the singleton level set has measure zero, so $g = 0$. Thus $\mathrm{spec}(\hat{x}) = [0,1]$, purely continuous spectrum. $\triangleleft$

Example 14.12 (Pure point spectrum). A diagonal operator $De_n = d_n e_n$ on ℓ^2 has each d_n as an eigenvalue, so its point spectrum is $\{d_n\}$. The full spectrum is the closure $\overline{\{d_n\}}$: a limit point of eigenvalues belongs to the spectrum even when it is not itself an eigenvalue. $\triangleleft$

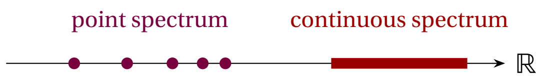

Figure 1: Two common parts of a self-adjoint spectrum on the real line: isolated eigenvalues (dots, possibly accumulating) and one common continuous-spectrum example (an interval with no eigenvectors). An interval is an example, not the definition of continuous spectrum.

Example 14.13 (Spectrum of the shift). The right shift on ℓ^2 has *no* eigenvalues, yet its spectrum is the entire closed unit disk $\{|\lambda| \le 1\}$ (a standard computation, stated here without proof). The gap between "eigenvalue" and "spectrum" is a purely infinite-dimensional effect. $\triangleleft$

Remark 14.14 (Bridge: the resolvent). The complement of the spectrum is the *resolvent set*, where $(T - \lambda\,\mathrm{id})^{-1}$ exists as a bounded operator. The resolvent is the analytic engine behind the spectral theorem. Optional Lecture 15 uses it to control how spectra and eigenstates respond to a perturbation.

Remark 14.15 (Residual spectrum and normal operators). For completeness, the spectrum splits into three parts. If $T - \lambda\,\mathrm{id}$ has a kernel, λ is in the *point spectrum*; if it is injective with dense range but no bounded inverse, λ is in the *continuous spectrum*; and if it is injective with range that is not even dense, λ is in the *residual spectrum*. The right shift exhibits the third kind: $\lambda = 0$ is residual, since S is injective but its range misses e_1 by a whole dimension. For *normal* operators—in particular every self-adjoint observable and every unitary—the residual spectrum is empty: if the range of $T - \lambda\,\mathrm{id}$ were not dense, its orthogonal complement $\mathrm{null}(T^\dagger - \overline{\lambda}\,\mathrm{id})$ would be nonzero, and for a normal operator that forces $\mathrm{null}(T - \lambda\,\mathrm{id}) \neq \{0\}$ (Lecture 9), putting λ in the point spectrum instead. This is why the observables of quantum mechanics carry only point and continuous spectrum.

Example 14.16 (Bridge: norm equals spectral radius). For a bounded self-adjoint (or normal) operator, $\|T\| = \sup\{|\lambda| : \lambda \in \mathrm{spec}(T)\}$ (a standard functional-analysis result, stated here without proof). A self-adjoint operator with spectrum in $[m, M]$ thus has norm $\max(|m|, |M|)$—the spectrum controls the size, just as for Hermitian matrices. $\triangleleft$

Example 14.17 (A computed spectrum). The operator $De_n = \frac{1}{n}e_n$ on ℓ^2 has eigenvalues $\{1/n : n \ge 1\}$ and spectrum $\{1/n\} \cup \{0\}$. The limit 0 lies in the spectrum—D is not invertible, since $De_n \to 0$—though it is not an eigenvalue. Its range is dense because it contains every finite-support sequence, so 0 is in the continuous spectrum. Nevertheless D has a complete eigenbasis and the pure-point spectral representation $D = \sum_{n\geq 1} n^{-1}|e_n\rangle\langle e_n|$; its PVM has atoms at $1/n$ and $P(\{0\}) = 0$. Thus continuous spectrum is not the same thing as a non-atomic part of the spectral measure. D is compact, with 0 its only spectral accumulation point. ◁

Example 14.18 (Spectrum of a multiplication operator). For multiplication by a continuous real g on $L^2[a,b]$, the spectrum is exactly the range (stated without proof),

$$\mathrm{spec}(M_g) = \overline{\{g(x) : x \in [a,b]\}},$$

self-adjoint because g is real. With $g(x) = x^2$ on $[-1,1]$ this is $[0,1]$, and there are no eigenvalues unless g is constant on a set of positive measure. ◁

3 Unbounded operators: position and momentum

The observables of quantum mechanics are typically unbounded. On $L^2(\mathbb{R})$ the *position* and *momentum* operators are

$$(\hat{x}\psi)(x) = x\,\psi(x), \qquad (\hat{p}\psi)(x) = -i\hbar\,\psi'(x),$$

each defined only on a dense domain (states for which the result is again in L^2). They are unbounded—the normalized sequence in Example 14.6 proves this directly for position. For momentum, fix a normalized $\phi \in \mathcal{S}(\mathbb{R})$ and use the wave-number parameter $k \in \mathbb{R}$: $\psi_k(x) = e^{ikx}\phi(x)$ remains normalized and belongs to $\mathcal{S}(\mathbb{R})$, while

$$\hat{p}\psi_k = \hbar k\,\psi_k + e^{ikx}\hat{p}\phi, \qquad \|\hat{p}\psi_k\| \geq \hbar|k| - \|\hat{p}\phi\| \longrightarrow \infty.$$

Thus momentum is unbounded as well. Both operators are self-adjoint on the right domains. For formal polynomial calculations below we use the Schwartz space $\mathcal{S}(\mathbb{R})$: it is dense in L^2 and invariant under multiplication by x, differentiation, and their polynomial compositions. Thus $\hat{x}\hat{p}$, $\hat{p}\hat{x}$, and the displayed powers are all meaningful there.

> **Theorem 14.19 (Canonical commutation relation).** On the common invariant dense domain $\mathcal{S}(\mathbb{R})$,
> $$[\hat{x},\hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar\ \mathrm{id}\,.$$

Example 14.20 (Deriving the commutator). For $\psi \in \mathcal{S}(\mathbb{R})$, both compositions remain in $\mathcal{S}(\mathbb{R})$. Apply the commutator: $\hat{x}\hat{p}\psi = -i\hbar\,x\psi'$ and $\hat{p}\hat{x}\psi = -i\hbar(x\psi)' = -i\hbar(\psi + x\psi')$, so

$$[\hat{x},\hat{p}]\psi = -i\hbar\,x\psi' + i\hbar(\psi + x\psi') = i\hbar\,\psi.$$

The $x\psi'$ terms cancel, leaving $i\hbar$ id. ◁

Example 14.21 (Physics bridge: momentum is diagonal in Fourier space). The plane wave $e^{ipx/\hbar}$ formally satisfies $\hat{p}\,e^{ipx/\hbar} = p\,e^{ipx/\hbar}$, but it is not in $L^2(\mathbb{R})$, so p is not an eigenvalue—it is continuous spectrum. The Fourier transform (Lecture 13) diagonalizes $\hat{p}$, turning $-i\hbar\,d/dx$ into multiplication by p; position and momentum are Fourier-conjugate multiplication operators. ◁

Example 14.22 (Optional extension: ladder operators). In natural units set $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$. Then $[a, a^\dagger] = \mathrm{id}$ follows from $[\hat{x},\hat{p}] = i\,\mathrm{id}$, and the number operator $N = a^\dagger a$ satisfies $[N, a^\dagger] = a^\dagger$, so $a^\dagger$ raises number by one. After stating the oscillator relation $H = \hbar\omega(N + \frac{1}{2})$, it also raises energy by one quantum $\hbar\omega$. ◁

Example 14.23 (Optional extension: polynomial commutators and dynamics). From $[\hat{x}, \hat{p}] = i\hbar$ and the Leibniz rule $[A, BC] = [A, B]C + B[A, C]$,

$$[\hat{x}, \hat{p}^2] = 2i\hbar\,\hat{p}, \qquad [\hat{p}, \hat{x}^2] = -2i\hbar\,\hat{x}.$$

These give the Heisenberg operator equations $\dot{\hat{x}} = \hat{p}/m$ and $\dot{\hat{p}} = -V'(\hat{x})$ for a particle in a potential. Taking expectations under the required domain hypotheses gives Ehrenfest's theorem. ◁

> **Proposition 14.24 (No finite-dimensional model).** There are no finite-dimensional matrices X, P with $[X, P] = i\hbar I$.

Proof. Taking traces, $\mathrm{tr}[X, P] = \mathrm{tr}(XP) - \mathrm{tr}(PX) = 0$ by cyclicity, while for dimension $n \geq 1$ and $\hbar \neq 0$, $\mathrm{tr}(i\hbar I_n) = i\hbar n \neq 0$. Thus finite matrices cannot satisfy the exact CCR. Continuous canonical degrees of freedom therefore require infinite dimension; this does not exclude genuine finite-level quantum models such as spins and qubits. ■

Remark 14.25 (Optional extension: the uncertainty principle). The commutator $[\hat{x}, \hat{p}] = i\hbar\,\mathrm{id}$ feeds directly into the uncertainty bound of Lecture 9: $\Delta x\,\Delta p \geq \frac{1}{2}|\langle[\hat{x}, \hat{p}]\rangle| = \frac{\hbar}{2}$. Heisenberg's inequality is the Cauchy–Schwarz bound applied to two non-commuting observables.

Example 14.26 (Optional extension: minimum-uncertainty Gaussians). The oscillator ground state $\psi_0 = \pi^{-1/4}e^{-x^2/2}$ (natural units) has $\langle\hat{x}\rangle = \langle\hat{p}\rangle = 0$ and, by direct integration, $\Delta x^2 = \frac{1}{2} = \Delta p^2$, so

$$\Delta x\,\Delta p = \tfrac{1}{2},$$

which is $\hbar/2$ once units are restored. This Gaussian is one minimum-uncertainty state. Translated, momentum-boosted, and squeezed Gaussians also attain the product; uniqueness would require extra centering, width, and ground-state conditions. ◁

Remark 14.27 (Domains matter). For unbounded operators, $\langle Tu, v\rangle = \langle u, Tv\rangle$ on a domain makes T *symmetric*, but *self-adjoint* additionally requires the adjoint to share that domain. Bounded normal operators also have a spectral theorem, but the real-observable PVM theorem and Stone-generator role used here require a genuinely self-adjoint realization; mere symmetry on a smaller domain is insufficient. Thus fixing the domain (and the boundary conditions) is part of defining an observable.

4 Compact operators and the discrete spectral theorem

The closest infinite-dimensional analogue of a Hermitian matrix is a *compact* self-adjoint operator.

> **Definition 14.28 (Compact operator).** An operator $T \in \mathcal{B}(\mathcal{H})$ is *compact* if it is a norm-limit of finite-rank operators: there are operators T_N with finite-dimensional range and $\|T - T_N\| \to 0$. Equivalently, T maps the unit ball to a set in which every sequence has a convergent subsequence—T repairs, on its image, exactly the compactness failure of Lecture 13.

A diagonal operator $De_n = d_n e_n$ is compact iff $d_n \to 0$ (truncate to the first N entries: $\|D - D_N\| = \sup_{n>N}|d_n| \to 0$), and the identity is *not* compact—the unit ball itself is not compact, as the orthonormal sequence of Lecture 13 showed. Compactness is thus a strong form of "finite-dimensional up to small corrections," and it buys back the full finite-dimensional spectral theorem.

Theorem 14.29 (Spectral theorem, compact self-adjoint case·cited). A compact self-adjoint operator T has a finite or countable set of nonzero real eigenvalues; each has finite multiplicity, and zero is their only possible accumulation point. Moreover

$$\mathcal{H} = \operatorname{null} T \oplus \overline{\operatorname{range} T},$$

the nonzero eigenspaces span $\overline{\operatorname{range} T}$, and any orthonormal basis of null T completes them to an eigenbasis of $\mathcal{H}$. In a separable infinite-dimensional space this basis may be listed as (e_n) with eigenvalues $\lambda_n \to 0$ (including zero entries from the kernel), giving $T = \sum_n \lambda_n |e_n\rangle\langle e_n|$. In finite dimension the corresponding list and sum are finite; no limiting assertion is needed.

This is Lecture 10's spectral decomposition with the kernel and separability qualifications visible. Standard confined Hamiltonians such as the box and oscillator are not compact—they are unbounded self-adjoint operators—but their resolvents are compact, which yields a discrete set of energy levels and an orthonormal basis of bound states.

Example 14.30 (Optional extension: a Hilbert–Schmidt integral operator). On $L^2[0,1]$ consider the integral operator

$$(Tf)(x) = \int_0^1 k(x,y) f(y)\, dy$$

with a square-integrable kernel k; it is compact (Hilbert–Schmidt). If k is Hermitian, $\overline{k(y,x)} = k(x,y)$, then T is self-adjoint, so Theorem 14.29 applies: the nonzero real eigenvalues form a finite/countable set with only possible accumulation at zero, and the kernel completes the eigenbasis. Because $L^2[0,1]$ is separable, that full basis can be listed countably. ◁

Example 14.31 (A spectral decomposition). In the separable case, a compact self-adjoint T with a listed eigenbasis acts by

$$Tv = \sum_n \lambda_n \langle e_n, v\rangle\, e_n,$$

the infinite-dimensional version of $T = PDP^{-1}$ (Lecture 6). Truncating the sum gives finite-rank operators converging to T in norm—how compact operators are handled in practice. ◁

Example 14.32 (Diagonalizing a compact operator). For a compact self-adjoint T with $Te_n = \lambda_n e_n$ and $v = \sum_n c_n e_n$,

$$Tv = \sum_n \lambda_n c_n e_n, \qquad \|Tv\|^2 = \sum_n |\lambda_n|^2 |c_n|^2, \qquad \langle v, Tv\rangle = \sum_n \lambda_n |c_n|^2.$$

Every quantity reduces to a weighted sum over the eigenbasis—exactly a diagonal matrix, now with infinitely many entries. ◁

Example 14.33 (Optional extension: the $\min(x,y)$ kernel). The integral operator on $L^2[0,1]$ defined by

$$(Tf)(x) = \int_0^1 \min(x,y)\, f(y)\, dy$$

is compact and self-adjoint, with eigenvalues $\lambda_n = \big((n-\tfrac{1}{2})\pi\big)^{-2} \to 0$. The decay of λ_n to 0 is the hallmark of compactness, and forces the spectrum to accumulate only there. ◁

Example 14.34 (Bridge: the number operator and compact resolvent). On ℓ^2 the number operator $N|n\rangle = n|n\rangle$ is self-adjoint with eigenvalues $0, 1, 2, \dots$ and the number states as eigenbasis. It is neither bounded nor compact (its eigenvalues run to ∞); but its resolvent $(N+1)^{-1}$, with eigenvalues $1/(n+1) \longrightarrow 0$, *is* compact—the usual route to a discrete spectrum. ◁

Example 14.35 (Bridge: resolvent of the number operator). The resolvent $(N+1)^{-1}$ acts on the number states by $|n\rangle \mapsto \frac{1}{n+1}|n\rangle$, with eigenvalues $1, \frac{1}{2}, \frac{1}{3}, \dots \longrightarrow 0$, so it is compact and self-adjoint with spectral decomposition $\sum_n \frac{1}{n+1}|n\rangle\langle n|$. Inverting a shifted unbounded operator is the standard way to expose a discrete spectrum. ◁

Example 14.36 (Physics bridge: particle in a box). For $-\frac{\hbar^2}{2m}\psi'' = E\psi$ on $[0, L]$ with $\psi(0) = \psi(L) = 0$, the solutions are $\psi_n(x) = \sqrt{2/L}\,\sin(n\pi x/L)$ with energies $E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}$. The ψ_n are the Fourier sine basis of $L^2[0, L]$ (Lecture 13). The Hamiltonian is unbounded, but its resolvent is compact, producing the same discrete eigenbasis shape as Theorem 14.29. ◁

Example 14.37 (Physics bridge: the harmonic oscillator). The oscillator Hamiltonian $H = \hbar\omega(N + \frac{1}{2})$ has spectrum $\{\hbar\omega(n + \frac{1}{2}) : n \geq 0\}$, a discrete ladder with eigenstates the number states $|n\rangle$. Built from the ladder operators $a, a^\dagger$, it is the canonical purely-discrete spectrum on an infinite-dimensional space. The Hamiltonian is unbounded; an appropriate shifted resolvent is compact. ◁

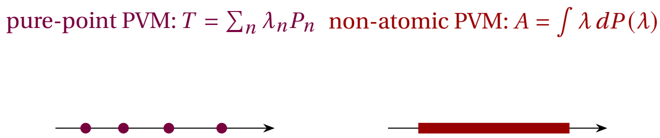

Figure 2: The PVM theorem is always an integral. It reduces to an eigenprojection sum for a pure-point spectral measure; a non-atomic spectral component needs a genuinely non-discrete integral. These measure types are not synonyms for the point/continuous spectrum classification of $A - \lambda I$: the compact diagonal $De_n = e_n/n$ has 0 in continuous spectrum but a pure-point PVM.

5 The spectral theorem and the postulates of quantum mechanics

For a general self-adjoint operator—bounded or unbounded—the universal form is a PVM integral. Whether it reduces to an eigenprojection sum is determined by the atomic or non-atomic character of the spectral measure, not merely by which points lie in the operator-theoretic continuous spectrum.

Theorem 14.38 (Spectral theorem, general self-adjoint case • cited). Every self-adjoint operator A has a *projection-valued measure* P on its spectrum with

$$A = \int_{\mathrm{spec}(A)} \lambda \, dP(\lambda).$$

Writing $\mu_\psi(\Omega) = \langle \psi, P(\Omega)\psi \rangle$, an unbounded A acts on the natural domain

$$D(A) = \{\psi \in \mathcal{H} : \int_{\mathrm{spec}(A)} \lambda^2 \, d\mu_\psi(\lambda) < \infty\}.$$

> For a pure-point spectral measure the integral reduces to the corresponding eigenprojection sum. A non-atomic spectral component, as for multiplication by x on $L^2[-1,1]$, requires a genuinely non-discrete integral. By contrast, $De_n = e_n/n$ has 0 in continuous spectrum but a pure-point PVM on the eigenvalues $1/n$, with no eigenprojection at 0. This general statement also covers unbounded pure-point operators, which need not be compact.

Remark 14.39 (Bridge: Stone dynamics). A self-adjoint Hamiltonian H generates the strongly continuous one-parameter unitary group $U(t) = e^{-iHt/\hbar}$. Conversely, Stone's theorem says every strongly continuous one-parameter unitary group has a unique self-adjoint generator. For unbounded H, the derivative formula $i\hbar \frac{d}{dt}U(t)\psi\big|_{t=0} = H\psi$ holds for $\psi \in D(H)$.

Example 14.40 (Measurement and the Born rule). Writing $A = \int \lambda\, dP(\lambda)$, the projection-valued measure P assigns to each spectral region Ω the projection onto the matching states. Measuring A in a normalized pure state $|\psi\rangle$ gives a value in Ω with probability $\|P(\Omega)|\psi\rangle\|^2$. In the ideal Lüders model, conditional on this probability being nonzero, the post-measurement state is $P(\Omega)|\psi\rangle/\|P(\Omega)|\psi\rangle\|$—the Born rule of Lecture 10, now valid for continuous spectra. ◁

Example 14.41 (Physics bridge: unitary time evolution). For $H|n\rangle = E_n|n\rangle$, Stone's theorem gives $U(t) = e^{-iHt/\hbar}$ acting by $|n\rangle \mapsto e^{-iE_nt/\hbar}|n\rangle$, so $\sum_n c_n|n\rangle$ evolves to $\sum_n c_n e^{-iE_nt/\hbar}|n\rangle$—the solution of the Schrödinger equation $i\hbar\,\partial_t\psi = H\psi$. ◁

Example 14.42 (Bridge: a two-level system). For a spin in a field, $H = \frac{\hbar\omega}{2}\sigma_z$ has eigenstates $|0\rangle, |1\rangle$ with energies $\pm\frac{\hbar\omega}{2}$. The state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ evolves to

$$\tfrac{1}{\sqrt{2}}\big(e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle\big),$$

whose Bloch vector precesses about the z-axis at frequency ω—Larmor precession, the simplest unitary dynamics. ◁

Example 14.43 (Bridge: spectral decomposition of a finite observable). $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$ is its own spectral decomposition: eigenvalues ± 1 with projections $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$. Measuring σ_z projects onto $|0\rangle$ or $|1\rangle$, and a function acts as $f(\sigma_z) = f(1)|0\rangle\langle 0| + f(-1)|1\rangle\langle 1|$. The finite-dimensional spectral theorem is the seed of this whole chapter. ◁

Example 14.44 (Bridge: a finite ideal measurement probability). Measuring σ_z in $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives each outcome probability $\frac{1}{2}$. Hence $\langle\sigma_z\rangle = \frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$—the average over the spectral projections. ◁

Example 14.45 (Bridge: a one-parameter group). The evolution operators satisfy $U(s)U(t) = U(s+t)$ and $U(0) = \mathrm{id}$, each $U(t) = e^{-iHt/\hbar}$ unitary and the group strongly continuous. For $\psi \in D(H)$, differentiation at $t = 0$ returns the generator, $i\hbar \frac{d}{dt}U(t)\psi\big|_0 = H\psi$. Stone's converse requires strong continuity; a merely algebraic one-parameter unitary group need not have such a generator. ◁

Example 14.46 (Physics bridge: the free particle). The free Hamiltonian $H = \hat{p}^2/2m$ has *continuous* spectrum $[0,\infty)$ and no normalizable eigenstates—its "eigenfunctions" are the non-L^2 plane waves. Time evolution $e^{-iHt/\hbar}$ still makes sense through the spectral integral, spreading a wave packet. Discrete and continuous spectra coexist in one theory. ◁

Remark 14.47 (Optional extension: two dynamical pictures). One may evolve states by $U(t)$ (the Schrödinger picture) or, equivalently, evolve observables by $A \mapsto U(t)^\dagger A U(t)$ (the Heisenberg picture). The two give identical predictions, related by a unitary—the same freedom as a time-dependent change of orthonormal basis.

Example 14.48 (Optional extension: Heisenberg equation of motion). In the Heisenberg picture an observable evolves by

$$\frac{d}{dt}A(t) = \frac{i}{\hbar}\,[H, A(t)],$$

the operator analogue of Hamilton's equations. For the free particle $[H, \hat{x}] = -i\hbar\,\hat{p}/m$ gives $\dot{\hat{x}} = \hat{p}/m$—the quantum statement that velocity is momentum over mass. ◁

Remark 14.49 (Functional calculus). For a bounded Borel function f, the spectral theorem gives a bounded operator

$$f(A) = \int f(\lambda)\, dP(\lambda).$$

For unbounded f, the same formula has natural domain $D(f(A)) = \{\psi : \int |f(\lambda)|^2 d\mu_\psi(\lambda) < \infty\}$. This defines evolution and band projections; for z in the resolvent set it defines the bounded resolvent $(A - z\,\mathrm{id})^{-1}$. Functions of operators, introduced for matrices in Lecture 6, reach their domain-aware form here.

Remark 14.50 (Optional extension: mixed states). The most general state is a *density operator*: a self-adjoint, positive, trace-one ρ. For bounded A, $\langle A\rangle = \mathrm{tr}(\rho A)$. For an unbounded observable this formula requires the appropriate finite spectral expectation and integrability/domain hypotheses; it can diverge for a valid density operator. Pure states are the rank-one projections $|\psi\rangle\langle\psi|$; the reduced state of an entangled pair (Lecture 12) is a genuinely mixed ρ.

Example 14.51 (Optional extension: expectation as a trace). For the qubit state $\rho = \frac{1}{2}(\mathrm{id} + \vec{r}\cdot\vec{\sigma})$ with Bloch vector $\vec{r}$,

$$\langle\sigma_z\rangle = \mathrm{tr}(\rho\,\sigma_z) = r_z,$$

so the Bloch coordinates are exactly the expectations of the Pauli observables. A pure state has $|\vec{r}| = 1$ (the Bloch sphere's surface); a mixed state lies strictly inside. ◁

Example 14.52 (Optional extension: a finite collapse calculation). Measuring σ_z on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ gives $+1$ with probability $|\alpha|^2$. Conditional on observing $+1$ with $\alpha \neq 0$,

$$|\psi\rangle \;\longmapsto\; \frac{P_+|\psi\rangle}{\|P_+|\psi\rangle\|} = |0\rangle,$$

where $P_+ = |0\rangle\langle 0|$ is the spectral projection for eigenvalue $+1$. A repeat measurement now returns $+1$ with certainty—the projection postulate. ◁

Remark 14.53 (The layered quantum dictionary). The core model uses a normalized vector as a representative of a pure-state ray, a self-adjoint observable, an ideal PVM/Lueders measurement instrument, and a strongly continuous unitary group for closed-system dynamics. The general replacements are density operators for states, POVMs together with instruments for measurements, and quantum channels for open dynamics. Finite-level systems are genuine quantum mechanics; infinite-dimensional Hilbert space is needed for continuous canonical degrees of freedom and scattering.

Remark 14.54 (Why infinite dimensions). The trace obstruction of Proposition 14.24 is no technicality: the canonical commutation relation, the continuous spectra of position and momentum, and the scattering states of the free particle all *require* an infinite-dimensional Hilbert space. This does not exclude finite-level quantum systems such as spin and qubits; it identifies the extra scope required by continuous canonical degrees of freedom.

Summary of main concepts

On a complex $\mathcal{H}$, bounded operators have adjoints and point, continuous, and residual spectrum. Domains govern unbounded position and momentum; their exact CCR has no finite-matrix model. Compact self-adjoint operators have a kernel-completed eigenbasis and nonzero spectrum accumulating only at zero. General self-adjoint operators use domain-aware PVMs, and strongly continuous dynamics use Stone generators. The pure/projective/closed QM model sits inside density-operator, POVM/instrument, and channel formulations.

- **Bounded operator**—finite operator norm = continuity; adjoint via Riesz; the self-adjoint, unitary, and normal classes persist.
- **Spectrum**—failure of bounded invertibility in $\mathcal{B}(\mathcal{H})$; point, continuous (dense range), and residual (non-dense range), with no residual spectrum for normal operators.
- **Unbounded observables**—$\hat{x}$ and $\hat{p} = -i\hbar\, d/dx$ obey $[\hat{x}, \hat{p}] = i\hbar\,\mathrm{id}$, impossible in finite dimensions (a trace argument).
- **Compact self-adjoint**—nonzero eigenvalues have finite multiplicity and only accumulate at zero; kernel plus nonzero eigenspaces gives an eigenbasis.
- **General spectral theorem**—$A = \int \lambda\, dP(\lambda)$ on its natural domain; pure-point PVMs reduce this to eigenprojection sums, while non-atomic components do not; strongly continuous unitary groups have self-adjoint Stone generators.
- **QM dictionary**—pure/PVM/closed is the core model; density operators, POVMs with instruments, and channels are the general replacements.

Exercises for this lecture are in the companion sheet `exercises-14.pdf`.